VII.

Reference solution 1：

The following equations

$$F = ma\ , \tag{1}$$

$$a = \frac{v^2}{r}\ , \tag{2}$$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}\ , \tag{3}$$

give the electron's total energy:

$$U = \frac{-e^2}{8\pi\varepsilon_0 r_0}\ . \tag{4}$$

The condition $E_B = -U = 13.6\text{eV}$ gives hydrogen's orbital radius and speed:

$$r_0 = 5.29\times10^{-11}\text{m}\ , \tag{5}$$

$$v_0 = 2.19\times10^{6}\text{m/s}\ . \tag{6}$$

The Larmor formula gives the initial radiated power:

$$P = \frac{e^2a^2}{6\pi c^3\varepsilon_0} = \frac{e^6}{96\pi^3c^3\varepsilon_0^3m^2r_0^4}\ . \tag{7}$$

During a short interval Δt, radiation reduces the electron's total energy U by

$$\Delta U = -P\times\Delta t\ , \tag{8}$$

$$\Delta U = \frac{-e^2}{8\pi\varepsilon_0}\left(\frac{1}{r-\Delta r}-\frac{1}{r}\right) = \frac{-e^2}{8\pi\varepsilon_0 r^2}\Delta r\ , \tag{9}$$

Here Δr is the decrease in orbital radius. The relation between time and radius r is

$$\Delta t = \frac{12\pi^2c^3\varepsilon_0^2m^2r^2}{e^4}\Delta r = A\times4\pi r^2\Delta r\ , \tag{10}$$

where $A = \dfrac{3\pi c^3\varepsilon_0^2m^2}{e^4}$.

For a sphere of radius r_0, $4\pi r^2\Delta r$ is the volume of a thin shell at radius r. Summing from r_0 to 0 gives volume $\dfrac{4}{3}\pi r_0^3$. Correspondingly, the electron orbit shrinks from r_0 to 0 in a time